

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network
Security COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I G. Dhanusha Sree ⁵⁰¹² (19232) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Dhanusha Sree

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them. Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

Develop a concept map comparing
Substitution techniques:-

Substitution ciphers are classical encryption techniques in which plaintext characters are replaced with other characters or symbol.

Concept map

Substitution Techniques

replace plaintext with
different symbols or characters.

Caesar cipher
Low complexity



Vigenere cipher
Repeating key used

Monoalphabetic cipher
Moderate complexity



Hill cipher
Invertible matrix
Key

Comparison of Substitution Techniques :

Key Usage → Fixed Shift

Encryption → Single letter shift

Complexity → Very low

Key space → 25 shifts

Explanation of Each technique

- * Caesar cipher

- * Monoalphabetic cipher

- * Vigenere cipher

- * Hill cipher

Key usage, complexity and security:-

caesar uses one numerical shift

caesar is simplest, monoalphabetic.

caesar and monoalphabetic ciphers are vulnerable to basic frequency.

Security Analysis:

Among these four classical techniques, Vigenere and Hill offer more protection against basic frequency analysis than Caesar and monoalphabetic ciphers.

Conclusion:

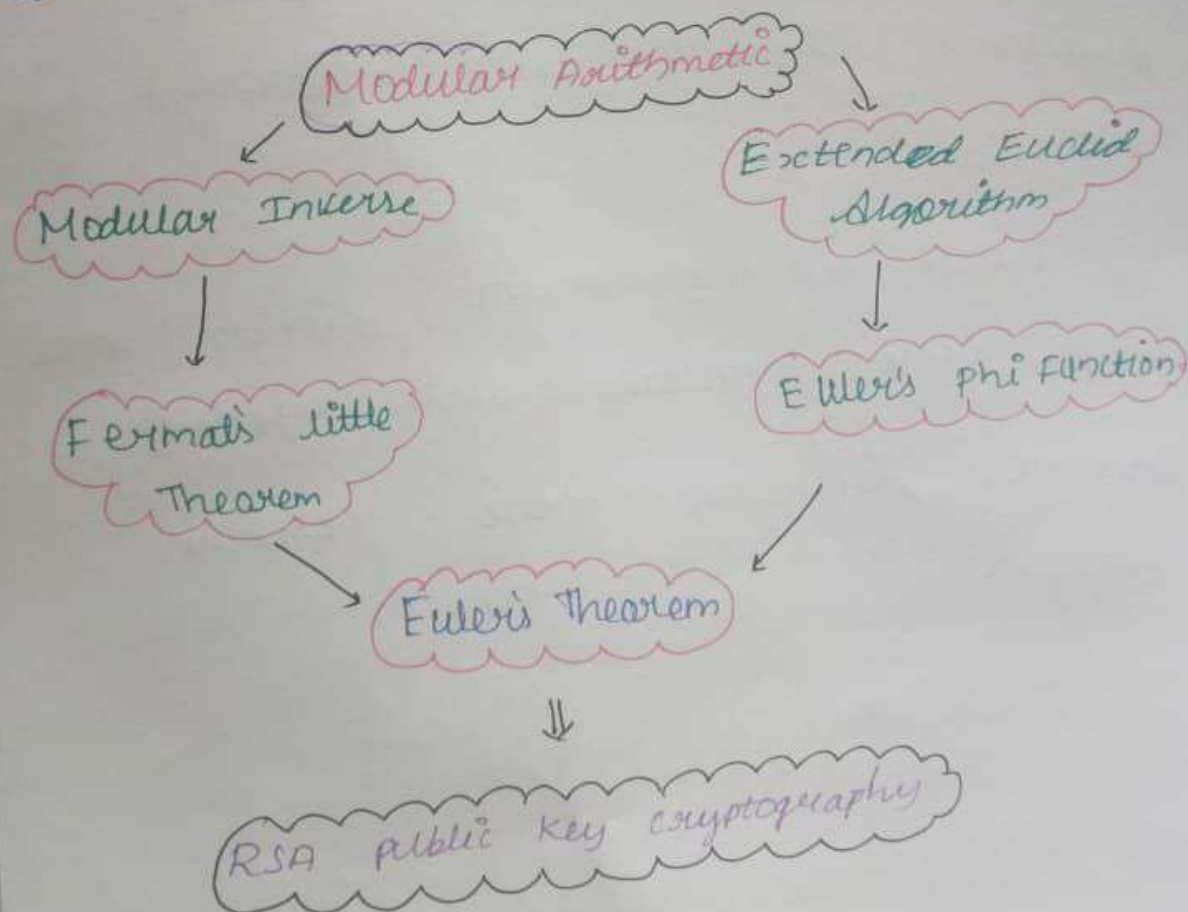
Substitution techniques replace plaintext characters using different key mechanics.

Caesar and monoalphabetic ciphers are simple but vulnerable, while Vigenere and Hill introduce more complex encryption.

57 Concept Map of Number Theory concept used in cryptography :-

Number theory is an important mathematical foundation of cryptography.

It uses properties of integers, prime numbers, divisibility and modular arithmetic to create encryption and decryption algorithms.



Modular Inverse

The modular inverse of a number a modulo n is a number x

$$ax \equiv 1 \pmod{n}$$

A modular inverse exists

$$\gcd(a, n) = 1$$

Extended Euclid Algorithm

The Greatest common divisor (GCD) of two integers and expresses

$$ax + by = \gcd(a, b)$$

Ex: Inverse of 7 modulo 26

$$26 = 3(7) + 5$$

$$7 = 1(5) + 2$$

$$5 = 2(2) + 1$$

Fermat's little Theorem:

if p is a prime number and p does not divide a , then

$$a^{p-1} \equiv 1 \pmod{p}$$

Euler's phi Function

Count the positive integers up to n that are relatively prime to n .

$$\phi(p) = p - 1$$

Connections Between the Concepts:-

- * Modular Inverse multiplicative inverse modulo n Calculate private Key
- * Extended Euclid Find Bezout co-eff and GCD Compute modular inverse

Most critical for public Key cryptography?

In RSA, Euler's phi function and Euler's Theorem establish the mathematical relationship that allows the encryption and decryption exponents to work together.

$$d \equiv e^{-1} \pmod{\phi(n)}$$

Conclusion:-

* Number Theory provides the mathematical foundation of public Key cryptography.

The Extended Euclid algorithm helps calculate modular inverse, Euler's phi function counts relatively prime numbers, Fermat's Little Theorem and Euler's Theorem establish RSA private exponent.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation